EEN025 Assignment 1

Group 6

September 17, 2026

Task 1

To maximize the net profit, we formulate the production problem as a linear program. The decision variables represent the annual production quantity of each model. Since each model has a different profit due to varing tax rates, material and labor costs, We need to calculate the the net profit per model, these values form the coefficients of the objective function.

$$\text{Material Cost}_i = \text{Price}_i \times \text{Material\%}_i \tag{1}$$

$$\text{Labor Cost}_i = \frac{\text{Salary}_i}{160} \times \text{Manhour}_i \tag{2}$$

$$\text{Pretax Profit}_i = \text{Price}_i - \text{Material Cost}_i - \text{Labor Cost}_i \tag{3}$$

$$\text{Net Profit}_i = \text{Pretax Profit}_i \times (1 - \text{Tax\%}_i) \tag{4}$$

Based on the (1),(2),(3),(4), the net profit per unit for each car model is computed as follows.

Table 1: Cost and profit comparison of different models

Model	**Panda**	**500**	**Musa**	**Giulia**
Material Cost	60.42	81.60	82.50	192.15
Labor Cost	5.00	3.09	4.75	16.25
Pretax Profit	40.58	51.31	62.75	218.60
Net Profit	**28.41**	**43.61**	**50.20**	**153.02**

Let x_i denote the number of units of car model i produced in the coming year, for $i \in \{\text{Panda}, 500, \text{Musa}, \text{Giulia}\}$.

The objective is to maximize the total net profit of the company. Therefore, we can write the function (5) with constraints .

$$\max \quad Z = c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 \tag{5}$$

$$\sum_{i=1}^{4} (\text{Material Cost}_i + \text{Labor Cost}_i)\, x_i \leq 4 \times 10^7 \tag{6}$$

$$x_1 \geq 120{,}000 \tag{7}$$
$$x_2 \geq 100{,}000 \tag{8}$$
$$x_3 \geq 80{,}000 \tag{9}$$
$$x_4 \geq 15{,}000 \tag{10}$$
$$\tag{11}$$

$$x_1 + x_3 \leq 300{,}000 \tag{12}$$

$$x_1, x_2, x_3, x_4 \geq 0 \tag{13}$$

The optimal production plan and the resulting maximum net profit are summarized in Table 2. The maximum achievable net profit is $Z^* = 2.4048 \times 10^7$ thousand Kr.

Table 2: Optimal production quantities

Model	**Optimal Production (units)**
Panda	120,000
500	100,000
Musa	80,000
Giulia	80,135

The results show that Panda, 500 and Musa are produced at the minimum quantities, while Giulia's production exceeds its minimum requirement.

This pattern reflects the relative profitability of each model. As computed in table 1, Giulia has the highest net profit per unit, higher than the rest models. Since the budget has limitation, this model allocates the minimum required production to the less profitable models and allocates the remaining budget to Giulia in order to maximize the net profit.

Budget The budget constraint's right-hand side can decrease to 2.643×10^7 thousand Kr. Below this threshold, it would no longer be feasible to satisfy all minimum production requirements. There is no upper limit, since any additional budget continues to be allocated entirely to Giulia.

Tax and Salary For Panda, 500 and Musa, the objective coefficient can decrease without limit without changing the optimal basis. However, if their net profit were to increase beyond the respective upper bounds , it would become optimal to produce more of them, changing the current solution structure. For Giulia, its net profit can increase without limit, but if it decreases below 119.90 thousand Kr, the optimal basis would change and the model would stop favoring maximal production of Giulia.

Task 2

In this task, the main objective is to determine the optimal production and shipping quantities to maximize the company's total net income while satisfying market demands.

We assume that the market-specific minimum demand requirements replace the minimum production requirements from Task 1, therefore, no separate minimum production constraints are used.

Since delivering vehicles to the US market incurs an export tax of 2.5% on the final selling price, whereas intra-Schengen deliveries incur no additional duties (14).

$$\text{Export tax}_{i,j} = \begin{cases} 0.025 \cdot \text{Price}_{i,j}, & \text{if } j = \text{US} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $P_{i,j}$ denotes the selling price of model i in country j.

To capture the multi-market distribution, the decision variables are defined over two indices: the car model $i \in \mathcal{I} = \{\text{Panda}, 500, \text{Musa}, \text{Giulia}\}$ and the destination country $j \in \mathcal{J} = \{\text{Poland}, \text{Italy}, \text{US}, \text{Sweden}\}$. Therefore, the pretax profit, net profit are shown in (15) and (16).

$$\text{Pretax Profit}_{i,j} = \text{Market Price}_{i,j} - \text{Total Unit Cost}_{i,j} - \text{Export Tax}_{i,j} \tag{15}$$

$$\text{Net Profit}_{i,j} = \text{Pretax Profit}_{i,j} \times (1 - \text{Tax}_i) \tag{16}$$

Where $\text{Total Unit Cost}_{i,j}$ denotes the sum of material cost, labor cost, and delivery cost for model i shipped to market j

Then, the net profit per unit for each car model to different countries is computed as table 3.

Table 3: Total cost, pretax profit, and net profit for each model-market combination

Model	Country	Total Cost (kkr)	Pretax Profit (kkr)	Net Profit (kkr)
Panda	Poland	66.220	19.780	13.846
Panda	Italy	65.420	40.580	28.406
Panda	US	68.920	77.330	54.131
Panda	Sweden	67.420	44.580	31.206
500	Poland	58.294	33.706	28.650
500	Italy	59.294	76.706	65.200
500	US	61.094	104.656	88.958
500	Sweden	59.894	90.106	76.590
Musa	Poland	99.250	0.750	0.600
Musa	Italy	87.250	62.750	50.200
Musa	Sweden	89.450	80.550	64.440
Giulia	Italy	208.400	218.600	153.020
Giulia	US	213.400	322.850	225.995
Giulia	Sweden	210.900	289.100	202.370

We can write the objective function and new constraint (18).

$$\max \quad Z = \sum_{(i,j) \in \mathcal{P}} R_{i,j} \cdot x_{i,j} \tag{17}$$

$$\text{subject to} \quad x_{i,j} \geq Dm_{i,j} \tag{18}$$

And we compute the solustions via Gurobi, and the results are shown in table 4.

Table 4: Optimal Production and Global Shipment Plan ($x_{i,j}$ in thousand-units)

Country	**Panda**	**500**	**Musa**	**Giulia**	**Total Market Intake**
Poland	75.00	20.00	10.00	—	105.00
Italy	35.00	40.00	80.00	8.00	163.00
US	40.00	252.64	—	3.00	295.64
Sweden	2.00	5.00	1.00	1.00	9.00
Total Produced	**152.00**	**317.64**	**91.00**	**12.00**	**572.64**

Note: Optimal Total Net Profit $Z^* \approx 3.623547 \times 10^{10}$ Kr.

All model-market shipments remain at their minimum demand levels except Fiat 500 shipments to the US. This route has the highest ratio of after-tax profit to budget expenditure, approximately 1.4561. After satisfying all minimum market demands, the remaining budget is therefore allocated to producing and shipping additional Fiat 500 vehicles to the US.

We assume that the budget covers production and transportation costs, while export tax is paid from sales revenue. Export tax is therefore excluded from the budget constraint but deducted when calculating net profit.

Task 3

Since foam is produced by mixing three components and the ration is 1:1:1, the maximum quantity of foam is determined by the component with the smallest total output.

Let $x_{i,j}$ denote the number of hours that production line $i \in \{1, 2, 3, 4\}$ spends processing component $j \in \{1, 2, 3\}$. The total quantity produced of component j across all lines is shown in (19).

$$T_j = \sum_{l=1}^{4} \text{rate}_{i,j}\, x_{i,j}, \qquad j = 1, 2, 3 \tag{19}$$

$$\text{Foam} = \min(T_1, T_2, T_3) \tag{20}$$

To transform (20) into a linear problem, we introduce an auxiliary decision variable F, representing the total quantity of foam produced. The linear program then maximizes F, subject to three upper-bound constraints that force F to be no greater than the total production of each component.

$$F \leq T_1, \tag{21}$$
$$F \leq T_2, \tag{22}$$
$$F \leq T_3. \tag{23}$$

When F is maximized subject to (21)–(23), the optimal value of F is forced to equal (20). Therefore, we can calculate the solution with Gurobi, and the results are shown in the table 5.

Table 5: Optimal line schedule and component production

Line	Component 1 (h)	Component 2 (h)	Component 3 (h)
1	0.00	100.00	0.00
2	88.28	61.72	0.00
3	80.00	0.00	0.00
4	0.00	53.79	146.21
Total production (units)	**2924.14**	**2924.14**	**2924.14**

The optimal solution allocates all available capacity across the four lines, with every line operating at its full capacity limit .The maximum production is shown in (24).

$$T_1 = T_2 = T_3 \approx 2924.14 \text{ units}, \tag{24}$$

which matches the maximum foam output $F^* = 2924.14$ units.

Task 4

Let I be the set of candidate locations. For each $i \in I$, let F_i denote the fixed opening cost, C_i the cost per unit of floor area, R_i the revenue per unit of floor area, and L_i, U_i the minimum and maximum sizes of an opened retailer. Floor area is measured in hundreds of square meters. Let W be the available investment budget and K the maximum number of retailers.

Define the decision variables:

$$y_i \in \{0, 1\},$$

where $y_i = 1$ if a retailer is opened at location i, and

$$x_i \geq 0,$$

representing its floor area in hundreds of square meters.

The MILP formulation is:

$$\begin{aligned}
\max \quad & \sum_{i \in I} R_i x_i \\
\text{subject to} \quad & \sum_{i \in I} (F_i y_i + C_i x_i) \leq W, \\
& \sum_{i \in I} y_i \leq K, \\
& L_i y_i \leq x_i \leq U_i y_i \qquad \forall i \in I, \\
& x_i \geq 0, \quad y_i \in \{0, 1\} \qquad \forall i \in I.
\end{aligned}$$

The budget constraint accounts for both fixed opening costs and size-dependent costs. The cardinality constraint limits the number of opened retailers. The linking constraints enforce $x_i = 0$ when $y_i = 0$, and $L_i \leq x_i \leq U_i$ when $y_i = 1$.

The objective maximizes total revenue. All constraints are linear. Since no numerical data are provided, numerical optimal locations, sizes, and revenue cannot be determined.

Task 5

All investment amounts are measured in millions of crowns. Assume that the entire budget of 1000 is invested, that duration and risk are investment-weighted averages, and that the 30

Let $x_i \geq 0$ be the investment in security $i \in \{A, B, C, D, E\}$. Introduce binary variables y_C, y_D, y_E, where a zero value prohibits investment in the corresponding security.

The after-tax return rates are:

$$r = (0.045,\ 0.054,\ 0.0357,\ 0.0308,\ 0.061).$$

To implement the strict inequalities, use positive margins $\varepsilon_d, \varepsilon_r, \varepsilon_A$. The MILP is:

$$\begin{aligned}
\max \quad & 0.045x_A + 0.054x_B + 0.0357x_C + 0.0308x_D + 0.061x_E \\
\text{subject to} \quad & x_A + x_B + x_C + x_D + x_E = 1000, \\
& x_B + x_C + x_D \geq 400, \\
& 9x_A + 15x_B + 4x_C + 3x_D + 2x_E \leq 1000(5 - \varepsilon_d), \\
& 2x_A + 3x_B + x_C + 4x_D + 5x_E \leq 1000(1.5 - \varepsilon_r), \\
& x_C \leq 1000y_C, \qquad x_D \leq 1000y_D, \\
& y_C + y_D \leq 1, \\
& x_E \leq 1000y_E, \\
& x_A \geq (1 + \varepsilon_A)y_E, \\
& x_i \geq 0 \quad \forall i, \\
& y_C, y_D, y_E \in \{0, 1\}.
\end{aligned}$$

The binary constraints prevent simultaneous investment in C and D. Positive investment in E forces $y_E = 1$, which requires investment in A above the one-million-crown threshold. All constraints are linear.

For $\varepsilon_d = \varepsilon_r = 10^{-4}$ and $\varepsilon_A = 0.001$, the optimal investments, in millions of crowns, are:

$$(x_A, x_B, x_C, x_D, x_E) \approx (227.245455,\ 0,\ 704.590909,\ 0,\ 68.163636).$$

The after-tax return is approximately 39.537923 million crowns. The weighted duration is 4.9999 years, the weighted risk is 1.4999, and approximately 70.4591

These margins deliberately tighten the original strict inequalities. If the average limits are instead interpreted as non-strict, the optimal return is approximately 39.538636 million crowns, attained at duration 5 and risk 1.5. Under the literal strict limits, this value is a supremum rather than an attained maximum.